

Soumil Iyer

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EDUCATION

Purdue University, West Lafayette, IN

Master of Science in Mechanical Engineering (GPA: 4.00)

January 2026 – December 2026 (Expected)

Bachelor of Science in Mechanical Engineering (GPA: 3.38)

August 2022 – December 2025

EXPERIENCE

Rapid Prototyping Lab Student Employee | Purdue University

August 2026 – Present

- Guide students from CAD/STL export through FDM/FFF printer/material selection and slicing in Bambu Studio and Cura
- Tune orientation, supports, infill, layer settings, and temperatures for each material to improve printability, strength, and finish
- Set up new printers and perform calibration, preventive maintenance, and troubleshoot existing units for reliable lab operation
- Inspect print setups before launch for failure risks and coach students on CAD/slicer changes that improve manufacturability

Graduate Teaching Assistant: Intro to Manufacturing | Purdue University

January 2026 – May 2026

- Instructing hands-on manufacturing and design labs to 19 students in lathing, milling, CNC, metal, wood, and electronics projects
- Taught assembly, tolerancing, and GD&T by applying ANSI/ASME Y14.5 standards to student drawings and manufactured parts
- Reviewed CAD drawings, BOMs, and assembly procedures against project specs to flag manufacturability and assembly issues

M.E. Undergraduate Research Intern | Purdue University

June 2025 – August 2025

- 3D-printed 60+ dogbone and 24 disk specimens in 6 thermoplastics in horizontal and vertical build orientations for Dr. Luz Sotelo
- Performed tensile testing (ASTM D638) and ultrasonic wave speed characterization to study orientation-dependent anisotropy
- Processed stress-strain curves and calculated Young's modulus and ultrasonic wave speed using MATLAB data analysis
- Produced formal technical documentation package demonstrating that build orientation significantly influences tensile modulus

M.E. Undergraduate Research Assistant: Energy Storage Analytics | Purdue University

August 2024 – December 2024

- Built a customtkinter Python app with file upload/analysis/plotting workflows to process 1,000+ battery datapoints for research
- Created reusable battery comparison/visualization workflows to streamline recurring Li-ion analysis and research documentation
- Interpreted battery datasets to connect performance trends with Li-ion cell design, behavior, and optimization considerations

PROJECTS

Mechanical Engineering Senior Design: Assisted Eating Utensil

August 2025 – December 2025

- Designed a sub-\$25 utensil using bearing-based passive self-leveling to damp tremor motion while preserving normal scooping
- Iterated CAD/3D-printed prototypes from user feedback, adding a ribbed grip, retention strap, and detachable spoon head
- Validated final prototype durability/usability with 2 m drop tests, repeated dishwasher cycles, and insertion/removal-force checks

Wind Tunnel Aerodynamic Analysis: Race Car Spoiler Lift and Drag Characterization

March 2025 – April 2025

- Designed/FDM-printed a SolidWorks race-car spoiler; tested 0–60 deg AoA in 10 deg increments at a constant 30.880 m/s
- Collected 3 trials/angle and processed 15-second LVDT datasets in MATLAB using averaging and standard-deviation error bars
- Identified maximum downforce/flow separation near 40 deg as C_l peaked ~ 0.65 before decreasing while C_d continued rising

Heat Transfer Analysis: Blanket Insulation Materials Comparison

March 2025 – April 2025

- Compared polyester, wool, and down under matched conditions using an epoxy skin analog, thermocouples, and LabVIEW
- Ran 1-hour cooling tests from 37 C toward ~ 5 C with 1-second LabVIEW logging; applied lumped-capacitance after $Bi \sim 0.03$
- Linearized cooling curves to calculate $k = 0.390, 0.0152, 0.0344$ W/m-K for polyester, wool, and down; wool performed best
- Compared conductivities with references, quantifying $\sim 2.63\%$, 67.66% , and 37.60% error while documenting uncertainty

CERTIFICATIONS

- Purdue University: Finite Element Analysis (2025); Programming in Python (2023), Programming with Arduino (2023)

SKILLS

CAD & Design: Siemens NX (PLM/PDM), SolidWorks, Fusion, CATIA, AutoCAD, Finite Element Analysis (FEA), MasterCAM

Manufacturing: CNC, 3D-Printing (FDM/FFF), Milling, Lathing, Soldering, Woodworking, GD&T (ANSI/ASME Y14.5)

Testing & Measurement: Tensile Testing (ASTM D638), Ultrasonic Testing (NDE), LabVIEW

Programming & Software: FANUC ROBOGUIDE, MATLAB, Python, C, Arduino IDE, Simulink, Minitab, Studio 5000

Manufacturing Systems: Lean Principles, Kaizen, 6S